

IoT driven Smart Hydroponic Farming system

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Abstract— The development of civilization brings traditional open-field agriculture into a major crisis because of decreasing property distribution per individual. This paper introduces a hydroponics monitoring system which applies Internet of Things technologies to regulate essential growth factors. The system executes its monitoring functions through NodeMCU (ESP8266) together with sensors that determine pH, TDS, temperature, humidity, light intensity, and water level. The system automates nutrient and water supply through the combination of pumps together with solenoid valves. The Arduino IDE analyzes live data which then gets displayed through cloud platforms. The paper focuses on developing a cost-effective solution which combines scalability and operational efficiency for indoor and constrained farming spaces.

Keywords— *Hydroponics, IoT, ESP8266, Precision Agriculture, Smart Farming.*

I. Introduction

Hydroponic cultivation combines Greek word roots that mean 'water' and 'labor' as it allows plant growth with nutrients dissolving in water instead of soil. The water and farmland constraints of urban areas make hydroponics an excellent sustainable cultivation system. The research project creates an intelligent hydroponic system with IoT sensors to control and enhance plant care operations especially when growing fenugreek plants.

The Food and Agriculture Organization reported in 2001 that the world population will exceed 9.7 billion people by 2050 which will require a 60% increase in global food output. The world faces an escalating problem of undernutrition because undernourished people now reach 11% of global population numbers. Food security has risen to become a critical issue in contemporary times which creates major problems for agricultural operations. Approximately 690 million people are malnourished. Global agricultural land functions primarily for livestock farming which requires double the territory than what plant-based agriculture needs. Each year the world loses about 30 percent of its entire food supply. Urban communities consist of numerous people throughout the entire world. The percentage of city-dwelling population reached 55% in 2018 while it was only 30% during 1950. The world population

living in cities will surge from the current 55% to anticipated levels of 68% during the next 30 years.

The accelerating pace of urbanization has not affected agriculture's ability to fulfill growing food requirements although this has involved substantial expenses. The agriculture sector presently needs greater amounts of energy and land along with enhanced water consumption and releases additional greenhouse gas emissions. The fundamental question exists regarding sustainable agricultural food delivery for urban settlements while simultaneously supporting poverty reduction across rural and urban territories.



The efficient reduction of poverty and urbanization processes requires major investments in smart farming systems above all other fields. The fundamental development of nations relies on agriculture because this sector has consistently triggered territorial disputes to become a cause of armed conflicts throughout history. Urban expansion creates increased resource competition between urban expansion and agriculture which diminishes agricultural production capabilities. Modern agricultural methods need to be deployed in order to attain efficiency goals and combat poverty and solve problems related to scarce land as well as climate change effects.

A. Hydroponics represents a contemporary solution in smart farming which deals with various issues that conventional agriculture faces. The water usage of hydroponics reaches 90% fewer levels than traditional farming methods as it delivers enhanced production through systematic nutrient management along with increased vertical crop arrangements. The photosynthetic photon flux density together with the shoot weight decreases along the vertical heights in Vertical Farming Systems (VFS) yet they generate higher crop yield per unit area than traditional horizontal hydroponic systems thus VFS emerges as an effective alternative.

B. Role of IoT in Hydroponics

IoT technology enables real-time monitoring together with automatic control functions and remote operation in hydroponic farming systems. Multiple sensors function to monitor real-time evaluation of pH, EC, temperature, humidity, light exposure and water level readings. The IoT system operates the pumps together with the lights along with fans and dosing systems to sustain perfect conditions for plant growth by processing data from sensors. Users control the system remotely by accessing the platform through dashboard applications which triggers alert notifications when systems deviate from normal operating parameters. The integration of IoT provides both real-time data logging capabilities which enable analysis alongside optimization opportunities and lets users pair the technology with AI solutions for predictive farming management. The implementation of IoT technology improves hydroponic farming operations through optimization of both efficiency and precision management systems and simplified operations.

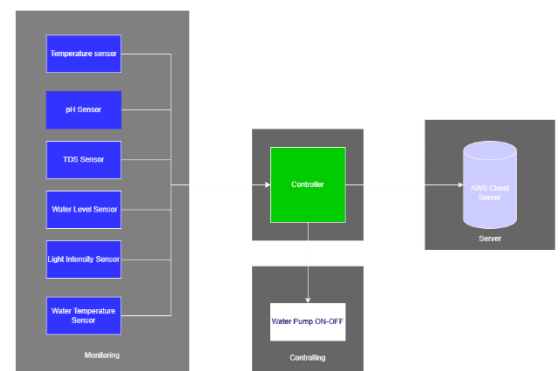
II. System Overview

The hydroponic system adopts advanced IoT technology to permit sustainable soil-free crop cultivation through efficient practices. The system employs various environmental sensors which monitor fundamental metrics such as temperature, humidity, pH, total dissolved solids (TDS) together with water level and water temperature and light intensity. The central ESP32 microcontroller receives constant sensor data that enables it to perform analysis to activate water pumps or lighting systems.

The system maintains cloud connection through MQTT protocol interactions that occur on platforms including AWS IoT Core. Real-time measurements get stored on the platform through its user-friendly mobile or web dashboard along with visual display capabilities and access points for users with diverse systems. Users benefit from notification alerts and they can conduct device manual operation as needed and inspect data trends.

Method	Medium Used	Description	Common Crops
Hydroponics	Nutrient-rich water solution	Plants grow in water mixed with essential nutrients, without soil.	Lettuce, herbs, spinach
Aeroponics	Air/mist	Roots are suspended in air and misted with nutrient solution.	Tomatoes, strawberries, basil
Aquaponics	Water (with fish waste)	Combines hydroponics and aquaculture; fish waste provides nutrients for plants.	Leafy greens, herbs
Cocopeat Farming	Coconut husk (coco coir)	Coco coir retains moisture and supports root growth. Nutrients added via water.	Tomatoes, cucumbers
Vermiponics	Water + vermicompost extract	Uses worm compost in water to feed plants.	Leafy greens, flowering plants
Grow Bag Culture	Inert medium (e.g., perlite)	Plants are grown in bags filled with inert material and fed with nutrient water.	Strawberries, peppers

III. IMPLEMENTATION



Monitoring Section :

The monitoring block involves collecting real-time data from various sensors. These sensors send data to the controller for processing and further action.

Sensors Used:

Temperature and Humidity sensor, pH Sensor, TDS Sensor, Water Level Sensor, Light Intensity Sensor, Water Temperature Sensor.

Controller Section

This is the central processing unit of the system. It receives data from the sensors, processes it, and takes decisions accordingly.

- Receives Data
- Processes Data
- Sends Data
- Controls Devices

Controlling Section

This block performs actions based on sensor data processed by the controller.

Water Pump ON/OFF:

water pump to add or circulate water and nutrients.

This can also be scheduled or triggered remotely.

Server Section

AWS Cloud Server

Stores and logs sensor data.

Provides remote access to data via a web or mobile app.

Can be used for analytics, reporting, and long-term monitoring.

IV. ROLE OF CLOUD

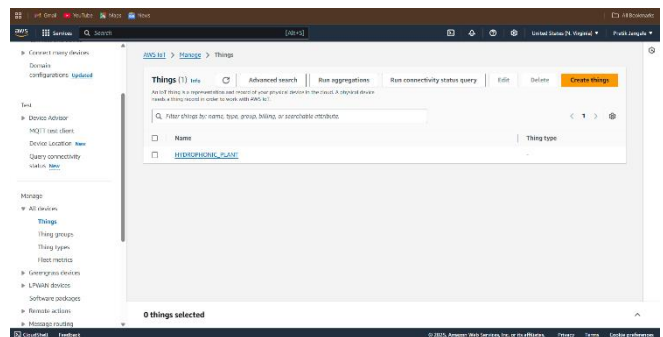
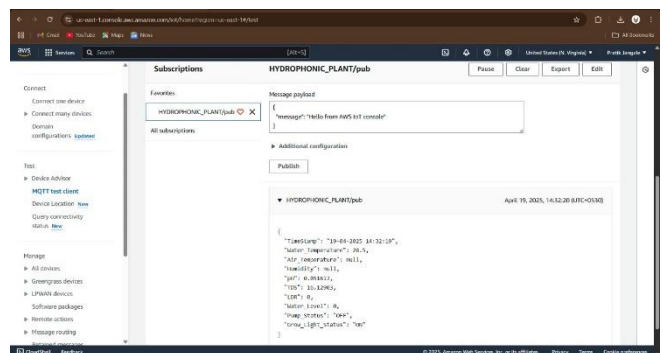
Feature	Description
Type	NoSQL database (key-value & document-based)
Scalable	Automatically scales with your data (no manual intervention needed)
Fast	Single-digit millisecond response time
Serverless	No server setup or maintenance
Highly Available	Data is replicated across multiple AZs (Availability Zones) in a region
Flexible Schema	No need to define rigid table schemas (you can have dynamic attributes)

Indexed Queries	Supports Primary Keys, Sort Keys, and Global Secondary Indexes
Time-to-Live (TTL)	Automatic data expiration feature
Streams	Capture data change events for triggering workflows
Security	Supports fine-grained access control via AWS IAM policies

Benefits:

You get a time-series log of all your sensor readings.

Useful for analysis, trends, reports, or even training ML models later.



V. RESULTS





An IoT-based hydroponic system reached real-time monitoring and automatic management of essential plant growth factors including pH, TDS, Temperature together with water temperature through its deployment in soilless farming systems. Continuous collection of accurate sensor data from pH and TDS meters combined with water level and temperature sensors along with light sensors was possible through the ESP8266 controller for remote cloud connectivity and automated pump control and improved watering efficiency. The agriculture system yielded healthful plants that grew more rapidly together with decreased operational work and resource utilization.

Underdeveloped regions can achieve maximum food production efficiency through hydroponics techniques which work in space-limited and poor-soil areas such as deserts. Systems that integrate AI functionality enable self-operated monitoring functions to continuously adjust nutrient levels as well as pH and environmental conditions for maximum growth results. When paired with artificial intelligence for image recognition plants can get diagnosed with diseases beforehand to enhance agricultural output and cut down crop destruction rate. Advanced technology development seeks to unite sustainable renewable energy systems with autonomous hydroponic units which create self-powered operations. Therapeutically inexpensive AI-controlled kits have the potential to enhance community food production thus diminishing dependence on imported food supplies. A combination of hydroponic farming with vertical agriculture methods alongside machine learning systems enables the